DS8001 Project Proposal: Permutation-Based Approximate kNN Search for Mixed-Type Hospital Encounters

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1. Problem and questions. Exact k-nearest-neighbor (kNN) search compares a query with all N indexed encounters. Using hospital encounters only as a mixed-type retrieval benchmark (no clinical claim), we ask: at matched recall@10, does a truncated MI-File reduce query latency and work relative to a truncated permutation scan (TPS), and how does the trade-off change with N? We also measure sensitivity to pivot count n, retained rank length L and reranking cap c.

2. Methods and scope. We implement exact brute-force kNN as ground truth, TPS, and a truncated MI-File. Both approximate methods use the same random pivots; each ranks pivots by distance (ties by pivot ID), retains its top L, and assigns absent pivots rank L+1. Both use Spearman-footrule candidate scores and true-distance reranking. TPS scores all N signatures; MI-File accumulates scores from posting lists for the query's retained pivots, so only objects sharing a retained pivot can enter its candidate set. A scikit-learn BallTree is an exact library control on a weighted-L1 embedding of the same distance; we will verify it against brute force, including boundary ties. Build uses O(Nn) pivot-distance evaluations plus ranking and stores O(NL) ranks. TPS query scoring is O(Nn); MI-File work depends on postings read and candidates. Each uses n query-pivot distances and reranks at most c candidates.

Distance. Six feature groups (age, utilization, admission/discharge, ICD-9 chapters, labs, medications) have equal total weight; features share their group's weight. Numeric/ordinal values use absolute differences after fixed index-fitted range scaling, without clipping; zero-range features are dropped. Nominal values use 0/1 mismatch. The selected numeric columns are complete in this dataset; categorical missing/unknown values and lab NOT_MEASURED are explicit fixed states, never pairwise-omitted. Thus the fixed weighted sum is a metric on the encoded feature states and has an exact weighted-L1 embedding for the BallTree. Out-of-index values retain their scaled differences.

3. Data and leakage control. Diabetes 130-US Hospitals (1999-2008): 101,766 encounters, 71,518 patients (Strack et al., 2014; UCI). Patient groups are split 90/5/5 into index, validation and test with a fixed seed; one query per patient is sampled. Scales and category vocabulary are fitted using index patients only; unseen categories map to OTHER. ICD-9 chapters use a fixed published grouping. Exclude encounter/patient IDs, readmitted, race, gender and weight.

4. Experiments and decision rules. Nested index sizes N=10k, 50k and the full index; 500 validation and 500 test queries; five pivot seeds; n in {32, 64}, L in {8, 16, 32}, c/N in {0.5%, 1%, 5%}. On validation, select Pareto configurations for recall@10 versus median latency; freeze the lowest-latency settings meeting recall targets 0.80 and 0.90 before test. Compare test latency and report distinct true-distance calls (reusing query-pivot distances), TPS signatures scored, MI-File postings/candidates, index bytes, build cost and peak memory separately. Report MI-File zero-overlap misses: true top-10 neighbors excluded before reranking. Also compare recall at matched true-distance-call budgets. Run single-threaded, warmed, repeated query batches on the same machine. Break candidate-score and exact-distance ties by encounter ID; report strict and boundary-tie-aware recall@10. Paired patient-level bootstrap intervals quantify query uncertainty; show results for each pivot seed to expose seed variability. An unreached recall target is reported as such, not extrapolated.

5. Validation and scope. We will test patient separation, metric properties, deterministic ties, and exact agreement of the L1 embedding and BallTree with brute force on sampled cases. The contribution is a controlled empirical comparison, not a new index. One dataset and one declared distance limit generalization; conclusions concern algorithmic retrieval, not clinical similarity. Python 3.11+; fixed seeds and experiment settings will be documented for reproduction.

References

Amato, G., & Savino, P. (2008). Approximate similarity search in metric spaces using inverted files. *Proc. InfoScale*.

Chávez, E., Figueroa, K., & Navarro, G. (2008). Effective proximity retrieval by ordering permutations. *IEEE TPAMI, 30*(9), 1647-1658.

Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. *JMLR, 12*, 2825-2830.

Strack, B. et al. (2014). Impact of HbA1c measurement on hospital readmission rates. *BioMed Research International*, 2014, 781670. UCI dataset, doi:10.24432/C5230J.